



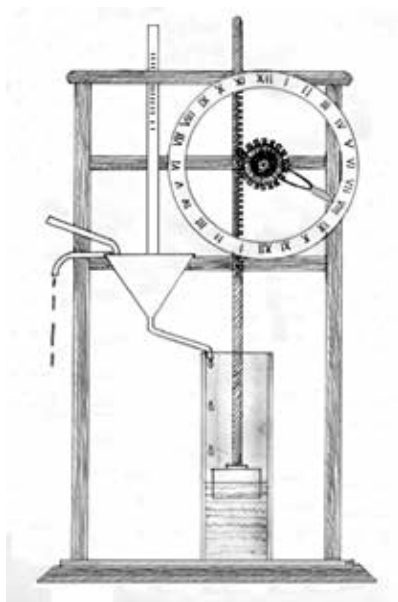
Netflix vs Apple

Netflix took to investors to list Apple and Amazon as rivals within the world of streaming entertainment and original content. <http://digit.in/NfxApp>



Blizzard games vulnerable

Severe vulnerability in Blizzard games allows remote attackers to run malicious code on gamers' computers. <http://digit.in/BlizHack>



A simple water clock

One of the most interesting accounts of a water alarm clock also comes from Greece. At Plato's academy, a vessel containing lead balls was gradually filled with water overnight at a steady rate, causing it to overflow near dawn. The balls would then crash into a copper platter resulting in a cacophony that would wake his students up.

Over the next 500 to 1500 years, water-based clocks were refined and used in several places around the world, including the Middle East and the Far East as well. Mechanisation and inclusion of gears and other clockwork components was done to increase the degree of accuracy. Similar clocks were also developed using incense sticks, candles, and perhaps the most popularised medium, thanks to its association with seafaring – sand.

A MECHANICAL CHANGE

As improved water clocks hit a ceiling even with the usage of mechanical components, the driving mechanism was replaced by weights that turned the gears and operated the automata within. During this time, the idea of a clock-face with dials indicating the hours and minutes also grew in popularity, whereas earlier mechanisations only involved bells striking the hour or smaller intervals.

This abandonment of liquid or other medium-based timekeeping for a purely mechanical process was truly brought about by the verge escapement mechanism in the now agreed time period of the 13th century.

Without going into a detailed explanation, it can be said that the verge escapement works on the basis of a circular wheel with jagged teeth creating an oscillating motion in another rod, the verge, that has two plates being nudged by the wheel at regular intervals. Clocks including the verge escapement increased accuracy to about ten seconds per day.

But the quest to make more accurate, intricate and compact clocks had by then being associated with superior craftsmanship. As a result, newer escapement mechanisms were being tried out regardless of the actual need for greater accuracy. One such mechanism brought into practice during the 15th century was the spring mechanism. But the overall precision of timekeeping was taken to a whole new level, and held there for a long time, by pendulum clocks in the 17th century.

Although Galileo had the idea of using a swinging bob to regulate timekeeping devices earlier in the 17th century, it is Christian Huygens who is credited as the inventor as he determined the formula that related pendulum swing to length and had the actual pendulum clock made in 1656. It was with further improvements, that accounted for things like the changes in the pendulum's length due to temperature variations, that the

accuracy of the pendulum clock was brought to a hundredth of a second a day and led to it being widely accepted as the standard for timekeeping until the arrival of quartz clocks.

THE NEW AGE

The first quartz clock was constructed in 1927 in Bell Laboratories. Based on the piezoelectric properties of the quartz crystal that made it vibrate consistently when electrified, early models of the quartz clock erred by less than a third of a second every year. While initially bulky and

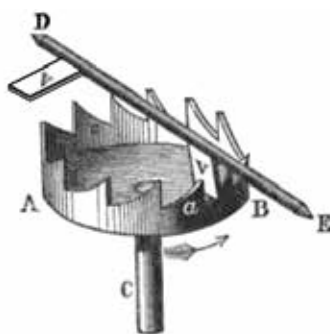


The NIST-7 was replaced in 1999 by the more accurate NIST-F1 as the official source for time in the United States

cumbersome, quartz crystals eventually showed up abundantly in wrist watches later in the 20th century.

When the first atomic clock was built in the 1950s, its accuracy was actually lower than quartz counterparts. Eventually, the first accurate atomic clock, based on a certain transition of the Caesium-133 atom, was built by Louis Essen in 1955 at the National Physical Laboratory in the UK. Atomic clocks are still widely regarded as the most accurate form of timekeeping.

Without going into the details of the atomic clocks improvement in the last 20 years, it suffices to say that they are capable of neither gaining nor losing a second in 20 million years! If that amazes you, consider this: the next challenger for the throne of the most accurate timekeeper, the optical clock, would only be a hundred seconds off if it were to be running for the entire existence of our universe. And with that, it is time to call it a day for this article.



The verge escapement mechanism – the ring turning in any particular direction would keep the verge oscillating